

NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
THEORY EXAMINATION
QUESTION PAPER

Month and Year of the Examination: August 07, 2026

Programme: B.Tech. (PIE)

Subject: **Heat Transfer**

Course No: PIPC-212

Semester: 4th

Time Allowed: 3 Hrs

Maximum Marks: 50

Note: Attempt five questions. Select one question each from section A, B, C, D. Section E (Question 9) is compulsory.

Any data book is not allowed. The relevant data is provided in the questions.

Section- A

1. Derive generalized heat conduction equation for the Cartesian coordinates. (10)
2. An exterior wall of a house may be approximated by 10 cm layer of common brick ($k = 0.75 \text{ W/m-deg}$) followed by 4 cm layer of gypsum plaster ($k = 0.5 \text{ W/m-deg}$). What thickness of rock wool insulation ($k = 0.065 \text{ W/m-deg}$) should be added to reduce the heat loss or gain through the wall by 75% for the same temperature difference on both sides of the wall? (10)

Section- B

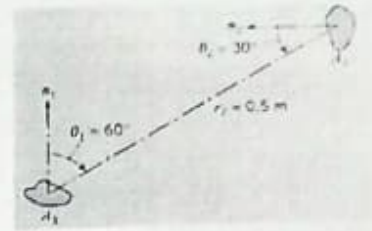
3. An infinitely long circular aluminium rod is attached at one end to a heated wall and transfers heat by convection to a cold fluid. If diameter of the rod is trippled, by how much would the heat removal change? And if the copper rod of the same diameter is used in place of aluminium, by how much would the rate of heat removal change. Thermal conductivity for Aluminium = 240 W/mK and for copper = 400 W/mK . (10)
4. Consider a concentric tube counter flow heat exchanger with hot and cold water inlet temperatures of 200°C and 35°C respectively. Hot and cold water outlet temperatures are 98°C and 87.2°C respectively. The flow rates of the hot and cold fluids are 42 and 84 kg/h, respectively. The overall heat transfer coefficient is $180 \text{ W/m}^2\text{K}$. Determine the maximum heat transfer rate that could be achieved for the given conditions. Hot Water $c_p = 4302 \text{ J/kg}\cdot\text{K}$; Cold Water $c_p = 4186 \text{ J/kg}\cdot\text{K}$. (10)

Section- C

5. Consider a steady, laminar, parallel flow of atmospheric air over a flat plate. The air has a temperature and free stream velocity of 300 K and 25 m/s respectively (a) Evaluate the boundary layer thickness at a distance of $x = 100 \text{ mm}$ from the leading edge. (b) If a second plate were installed parallel to and at a height of 3 mm from the first plate, what is the distance from the leading edge at which the boundary layer merger would occur? (c) Evaluate the surface shear stress at $x = 10 \text{ mm}$ from the leading edge. Air (300 K , 1 atm): $\rho = 1.161 \text{ kg/m}^3$, $\nu = 15.89 \times 10^{-6} \text{ m}^2/\text{s}$. (10)
6. An object of irregular shape has a characteristic length of $L = 1.0 \text{ m}$ and is maintained at a uniform surface temperature of $T_s = 400 \text{ K}$. When placed in atmospheric air at a temperature of $T_\infty = 300 \text{ K}$ and moving with a velocity of $V = 100 \text{ m/s}$, the average heat flux from the surface to the air is $20,000 \text{ W/m}^2$. If a second object of the same shape, but with a characteristic length of $L = 5 \text{ m}$, is maintained at a surface temperature of $T_s = 400 \text{ K}$ and is placed in atmospheric air at $T_\infty = 300 \text{ K}$, what will the value of the average convection coefficient be if the air velocity is $V = 20 \text{ m/s}$? (10)

Section- D

7. Plot and discuss the variation of spectral emissive power w.r.t. the wave length obtained by the Planck's distribution. (10)
8. Consider a small surface of area $A_1 = 10^{-4} \text{ m}^2$, which emits diffusely with a total, hemispherical emissive power of $E_1 = 5 \times 10^4 \text{ W/m}^2$. Determine (a) At what rate is this emission intercepted by a small surface of area $A_2 = 5 \times 10^{-4} \text{ m}^2$? (b) What is the irradiation G_2 on A_2 ? (10)



Section- E (Compulsory)

9. Answer any four:

- (a) Get the relation between coefficient of friction and Nusselt number using Reynolds analogy. (2.5)
- (b) Prove that the convective heat transfer coefficient is independent of length for the thermally fully developed flow in a duct. (2.5)
- (c) Write three major characteristics of a black body. (2.5)
- (d) Describe Wein's displacement law (2.5)
- (e) Define and briefly explain effectiveness of a heat exchanger. (2.5)
- (f) Derive critical thickness of insulation for a cylinder. (2.5)